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IARCO RESEARCH PROPOSAL

Effect of the Logical-Empirical Mapping (LEM) Scaffold on Project Scoping Time and Failure Recovery in Novice STEM Students

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Research Topic: STEM research-methods education: scaffolding novice project scoping

Title: Effect of the Logical-Empirical Mapping (LEM) Scaffold on Project Scoping Time and Failure Recovery in Novice STEM Students

Research Problem

First research projects in STEM rarely fail at the bench. They stall earlier, when a student has an interesting idea and no clear way to turn it into something one person can test in a few weeks. A student wants to know whether social media harms sleep, or whether AI tools improve learning. Weeks go into reading, and the questions that would make the idea testable are never settled: whose sleep, measured how, over what period. The project either grows past what can be finished or is quietly dropped.

The evidence for this is consistent, though small in scale. In an interview study of seven undergraduate novices, five found choosing a topic hard, and some said they did not know what makes a topic researchable [1]. Among 32 students at a science high school, identifying the research problem was described as one of the most daunting tasks, and writing the scope and limitations caused further difficulty [2]. The pattern reaches adults too: STEM teacher-researchers still list lack of research exposure and experience among their barriers [3]. Novices also report low confidence [1], so a stalled first project can be read by the student as proof that research is not for them.

The help that is offered mostly arrives once a topic already exists. The strategies suggested in [2] are seeking help, improving how sources are found and prioritising tasks. These are useful, but none answers the earlier question of how a large idea is cut down to a boundary that can be defended. This study addresses that step: the move from an abstract idea to a bounded, measurable design, before any data are collected.

Existing Literature

Scaffolding is the idea that a more capable other controls the parts of a task the learner cannot yet manage, so the learner can complete the rest [4]. A meta-analysis of inquiry-based learning reports that guidance improves learning activities, performance and outcomes [5], and a synthesis of 144 experiments found a positive average effect of computer-based scaffolding on cognitive outcomes in STEM ($g = 0.46$) [6]. These are learning outcomes inside a problem that has already been set, which is a different thing from deciding what to study.

How much structure to give is contested. Productive failure studies found that students who struggled with complex problems before instruction, and mostly failed to solve them, later outperformed students who received direct instruction first [7], [8]; the school studies in [8] involved Grade 7 mathematics. Critics argue that minimal guidance does not work for novices [9]. A recent case study of two biology teachers describes the scaffolding of student agency in inquiry classrooms and reports that one teacher's shift toward inquiry was driven by a willingness to take conceptual risks, while the other teacher followed procedure [10]. None of this work tests how much structure helps a student choose and bound a project. This study takes no side in the debate. It structures one step, boundary-setting, and leaves the problem-solving that follows unscaffolded, where room for productive struggle remains.

Several question-formulation tools already exist. The PICO format (population, intervention, comparison, outcome) came from clinical practice [11], and variants such as PICOS and SPIDER extend it to qualitative questions [12]. They prompt the user to fill slots. They do not say how the edge of a population is drawn or how an outcome is to be observed, so those decisions stay with the student, who is the person least able to make them.

First-order logic states exactly those decisions. A structure pairs a non-empty domain of discourse with an interpretation, predicates have a fixed arity, and truth is defined relative to the structure [13], [14]. In the sources consulted for this proposal, no study was found that uses these definitions as a planning aid for novice researchers. That is the gap. There is

evidence that novices struggle to bound projects [1]–[3], evidence that guidance helps once tasks are set [5], [6], question tools that name slots without boundary rules [11], [12], and exact definitions of domain, predicate and interpretation [13], [14]. These have not been tested together as a scaffold. The Logical-Empirical Mapping (LEM) architecture turns those definitions into an eight-stage sequence. This proposal uses it in two ways: as the scaffold whose effect is measured, and as the frame that structures the design of the study itself.

Research Question

Question. Does teaching the Logical-Empirical Mapping (LEM) architecture to novice STEM students (Grade 11–12 and first-year undergraduates, about 16 to 19 years old, recruited online) shorten the time to a bounded project scope and speed recovery from a first protocol failure, compared with standard research guidance?

Hypotheses. **H1:** students taught LEM will reach an approved project scope in fewer days than controls. **H2:** among students who record a protocol failure, those taught LEM will submit an adapted protocol in fewer days. The study measures these two respects, Project Scoping Velocity and Structural Failure Recovery Rate, and no others.

Key terms. A *novice* is a student starting a first independent project with no prior research experience. A *scope* is a written statement that defines the population and the property a project studies. A *protocol failure* is a logged error, instrument problem or unusable data run that forces a change to the plan.

Methodology

Two levels. The word methodology has two meanings in this proposal, and they are kept apart. At the first level, LEM is the intervention: the method novice students are taught for scoping their own projects. At the second level, LEM is the frame for the research design itself. Table 1 fills the eight coordinates of the architecture in order, each with its empirical realization. This is a proposal. It specifies what will be done and measured, and no data have been collected.

Table 1. The research design is built on the eight LEM stages.

Stage	Formal coordinate	Realization in this study
1. Universe	Domain of discourse D	Novice STEM researchers: Grade 11–12 students (or A-level/IB equivalent) and first-year undergraduates, about 16 to 19 years old, each starting a first independent project of 8 to 12 weeks. Excluded: anyone with prior lab work, co-authorship or research coaching, and anyone who has studied symbolic logic, model theory or discrete mathematics. Sixty students recruited online, 30 from each academic tier.
2. Predicate	Predicate P of arity n	Two two-place predicates. Scoped(s, t): student s holds an approved scope at day t. Recovered(s, t): student s has submitted an adapted protocol t days after a first failure. Arm(s, g) records the group, g being LEM or control.

3. Mapping rule	Interpretation I	*Project Scoping Velocity* is the number of days from a baseline idea statement to an approved scope. A scope is approved when the population (D) and the property studied (P) of the student's own project are both defined in writing and accepted by two raters blind to group. *Failure Recovery Rate* is the number of days from the first logged protocol failure to an adapted protocol accepted by the same raters. Log entries and a short interview after a failure are coded by thematic analysis [15], asking whether the student reads the error as personal failure or as noise (ε).
4. Statement	Well-formed formula	H1: days to an approved scope are lower in the LEM group. H2: days from a first failure to an adapted protocol are lower in the LEM group. Both are comparative claims about the two measured respects. A research hypothesis is not automatically a first-order formula, so they are stated in words.
5. Noise	Assignment g; error model Y = H(X) + ε	Observed days differ from true days because of self-logging, rater judgement and uneven school calendars (ε). Both groups get the same deadlines, tools and feedback schedule, timestamps are checked against document history, raters work blind to group, and rater agreement is reported.
6. Elements	Extension of P; dataset as structure	For each student, two counts of days and the coded log entries, stored under participant codes. A dataset is not a model by default, so a data dictionary fixing the domain, the variables and their interpretations is written before collection. Recruitment takes four weeks, scoping is observed for 42 days, and projects run to week 12.
7. Verdict	Tarskian satisfaction (not statistical inference)	The between-group difference in days is estimated with an effect size and a 95% confidence interval. Time-to-event methods [16] treat students who have not yet reached the event as censored rather than dropped. A p-value is an inferential quantity, not a truth value. With 30 students per group the study is sized to detect large effects, and smaller ones will be reported as imprecise.
8. Across realities	Modal (possible-worlds) semantics [17]	The two academic tiers are two settings in one study, and the tier comparison is exploratory. Replication in other cohorts, countries and disciplines is left as follow-up work, because possible-world semantics is a formal frame and not a replication.

Intervention and control. LEM is what the experimental group is taught. Video modules show students how to fix D, P and I for their own project (Stages 1 to 3) and how to use the error model of Stage 5 to treat a failed run as noise and not as a verdict on themselves. As an example, a student with the idea "phones hurt sleep" fixes D as Grade 11 students at one school who own a smartphone, P as a relation between a student and nightly hours of sleep, and I as the hours between logged bedtime and waking in a seven-day diary. The control

group receives standard institutional research manuals in modules of matched length, on the same platform, with the same deadlines and weekly written feedback. Students are allocated at random within each tier, 15 per group, so a difference in the two respects can be attributed to the guidance and not to who chose it. Stages 6 to 8 of LEM are not taught, because they come after data exist.

Boundaries of the design. Students cannot be blinded to the guidance they receive, so blinding of raters is used to limit the effect on outcomes. The groups could share material, so they use separate course spaces, and an exit question records any access to the other group's materials. Volunteers are probably more motivated than typical novices, so results describe motivated online STEM novices and not teacher-led classrooms. Only two respects are measured. The study does not test the later stages of LEM, and it does not claim that research projects are first-order structures: the correspondence is structural, not an identity.

Ethics. Adults sign an electronic consent form. Minors need a parent or guardian's signed consent and their own assent before access. Names, emails and institutions are replaced by participant codes, the key is kept in an offline encrypted file, and interview recordings are deleted after transcription. The protocol will be submitted to an ethics committee or an equivalent independent reviewer before recruitment, and the control group receives the LEM materials once data collection ends.

Research Topic

The topic is timely because youth research programmes and competitions bring in students with no previous project experience, and the studies reviewed here document scoping difficulties in exactly that group [1]–[3]. The tool proposed costs almost nothing to deliver. It needs a free course platform and a few short videos, so a positive result could be adopted by schools, mentors and competition organisers without new resources.

It is also new in a specific way. It takes definitions that sit in mathematical logic texts [13], [14] and tests them as a practical scaffold for the least structured stage of research. The study is clear about what would count as failure. If LEM does not shorten scoping, or does not speed recovery from a failed run, the result will show that a formal scaffold is not enough at this stage, which is useful to anyone designing guidance for novices. If it works, the next steps are clear: extend the test to Stages 6 to 8 and to teacher-led classes, and compare LEM directly with PICO-style prompts.

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